



ASMedia ASM2464PD

Improvement of the VBUS Peak Current



Disclaimer

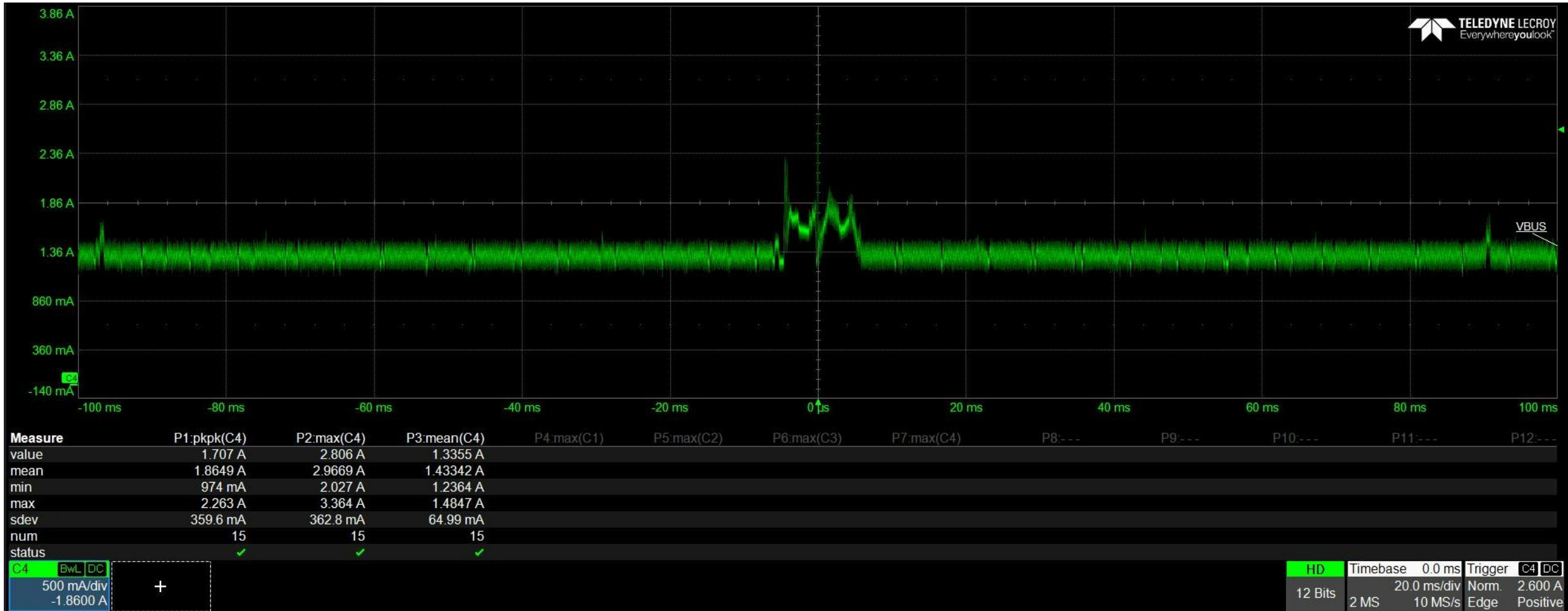
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- Large capacity NVMe SSD (i.e. 2TB/4TB, etc.) could consume large power while performing data read/write access and lead to ASM2464PD VBUS current peak larger than 3A.
- To minimize the peak current on VBUS, additional capacitance (e.g. >900uF) would be required on 3.3V power for M.2 SSD.
- It is strongly recommended that sufficient capacitors are implemented with low connection inductance between power and ground planes when capacitors are mounted on either top or bottom.



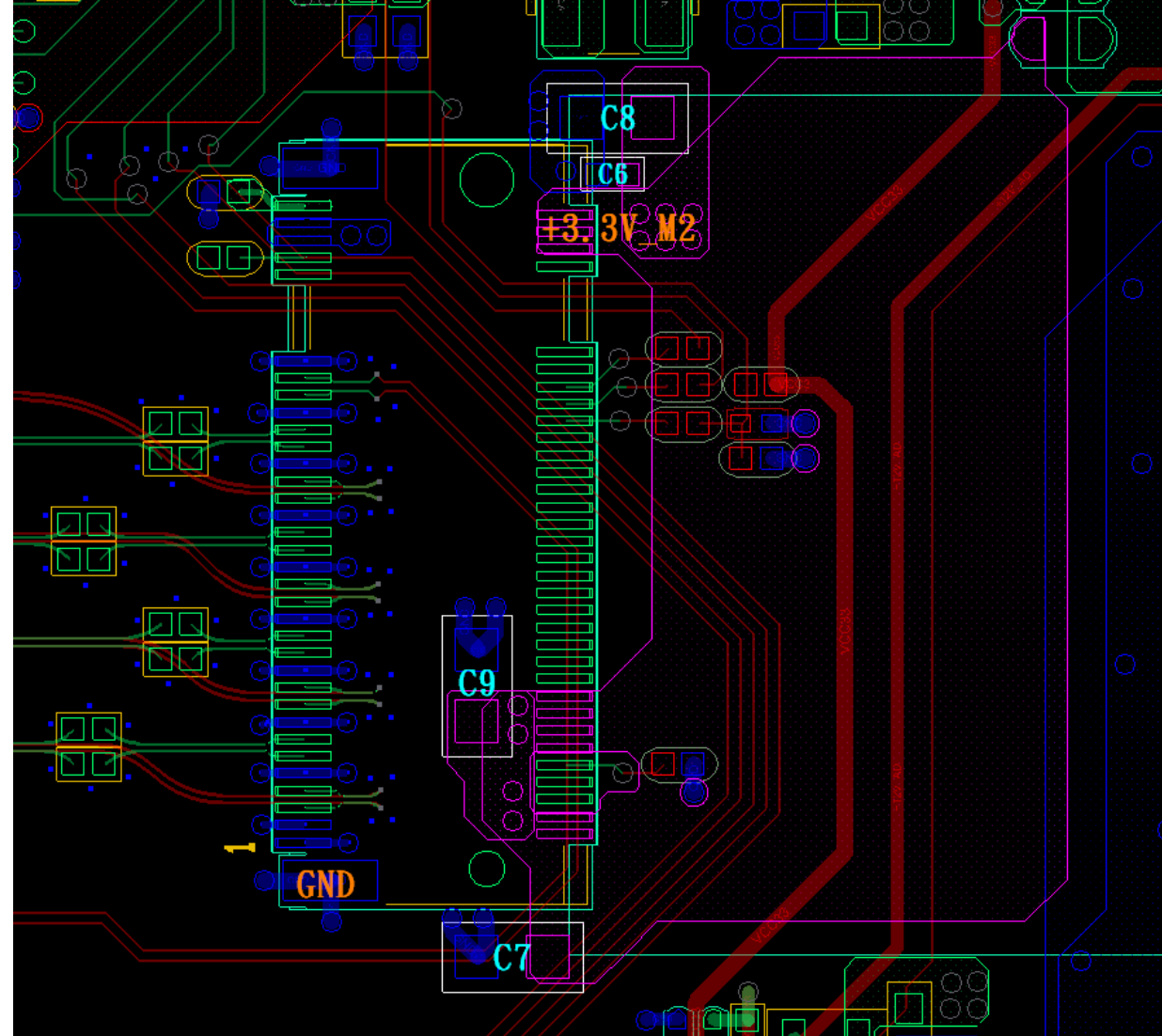
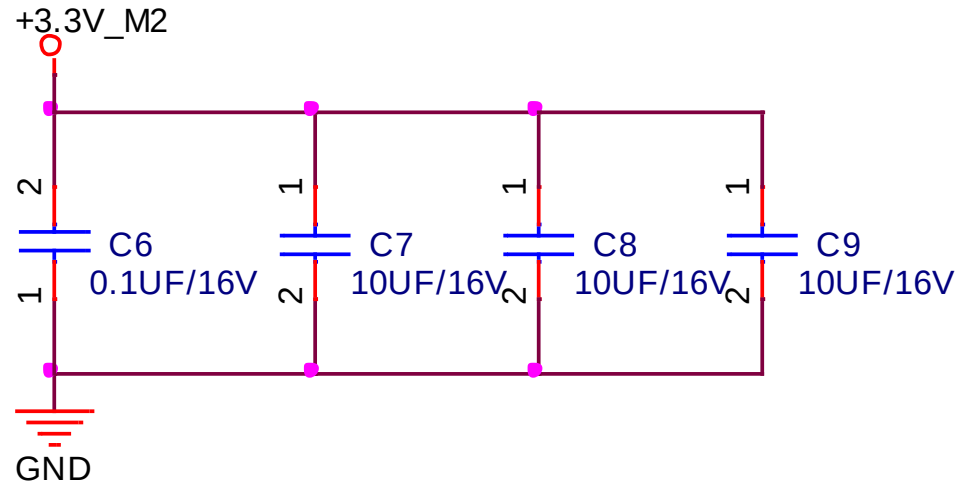
Original USB4/TBT3 VBUS Current (Power Consumption when SSD is Samsung 990PRO 2TB)

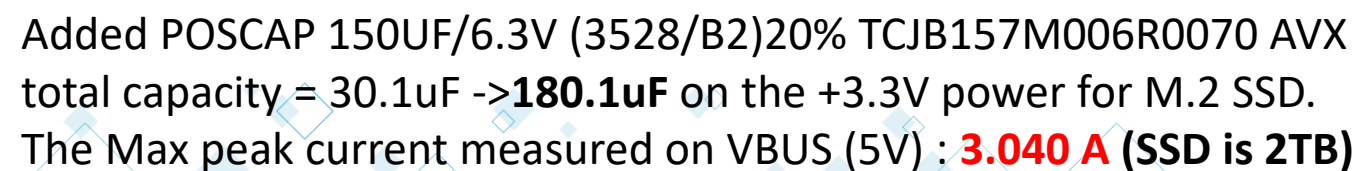


- Original Current measured on VBUS (5V) when 2TB SSD is connected. The original capacitance is **30.1uF** on the +3.3V power for M.2 SSD .
- The Max VBUS peak current measured: **3.364A** (SSD is 2TB)



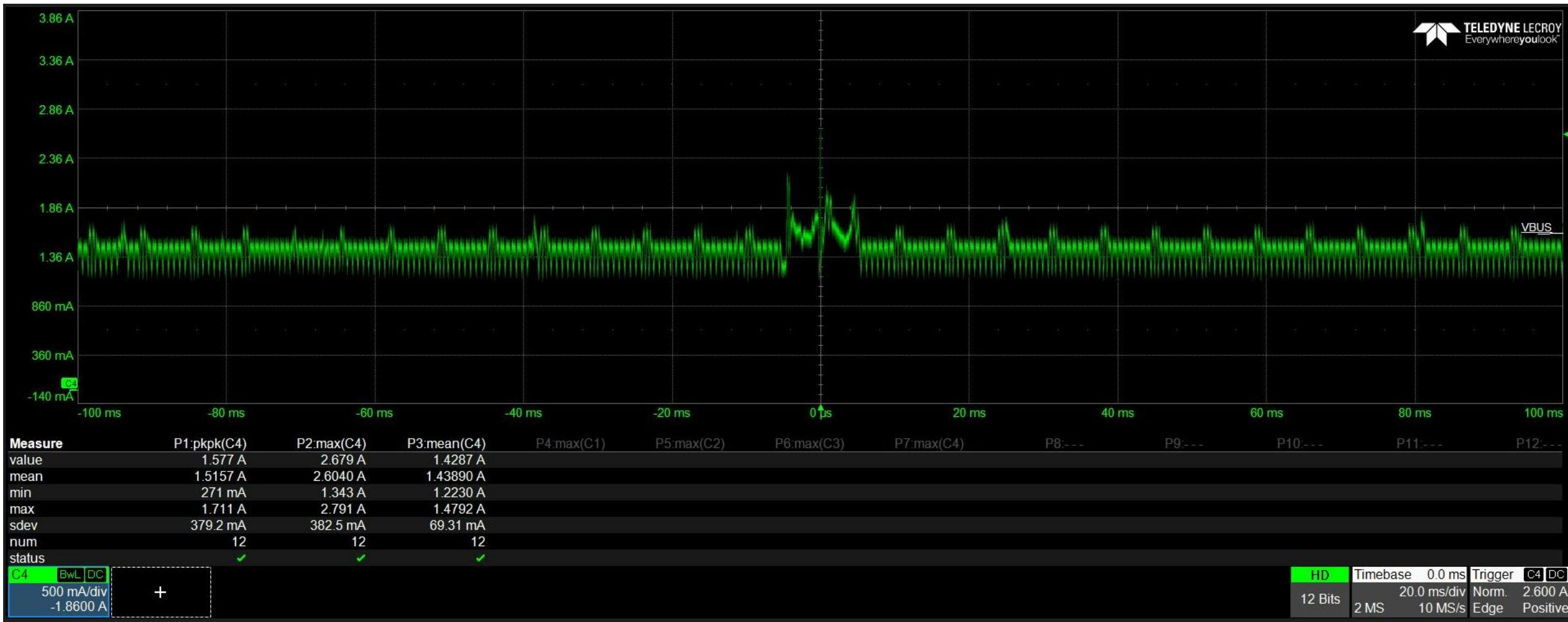
Original +3.3V power capacitors for M.2 connector







Improved USB4/TBT3 VBUS Current by Adding capacitance to **330.1uF** On +3.3V for M.2

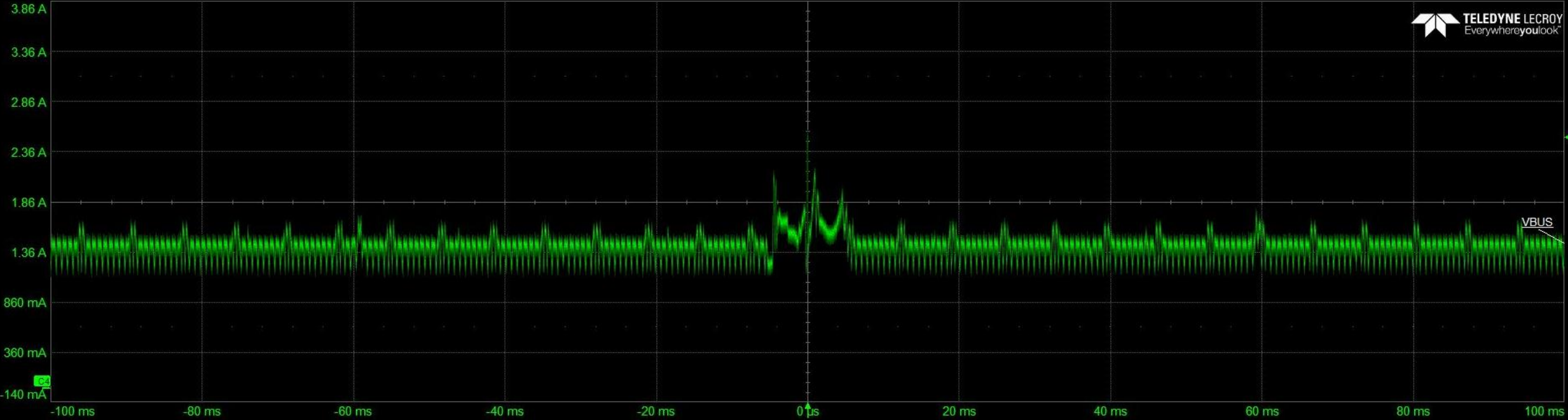


Added POSCAP 300UF/6.3V (3528/B2)20% TCJB157M006R0070 AVX ,
total capacity = 30.1uF -> **330.1uF** on the +3.3V power for M.2 SSD.
The Max peak current measured on VBUS (5V) : **2.791 A** (SSD is 2TB)



Improved USB4/TBT3 VBUS Current by Adding capacitance to **480.1uF** on +3.3V for M.2

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Measure	P1:pkpk(C4)	P2:max(C4)	P3:mean(C4)	P4:max(C1)	P5:max(C2)	P6:max(C3)	P7:max(C4)	P8:---	P9:---	P10:---	P11:---	P12:---
value	1.458 A	2.565 A	1.4084 A									
mean	1.3697 A	2.4184 A	1.3599 A									
min	192 mA	934 mA	839.7 mA									
max	1.604 A	2.663 A	1.4549 A									
sdev	420.0 mA	526.5 mA	185.2 mA									
num	9	9	9									
status	✓	✓	✓									

C4 BwL DC

500 mA/div

-1.8600 A

+

HD

12 Bits

Timebase 0.0 ms

20.0 ms/div

2 MS

10 MS/s

Trigger C4 DC

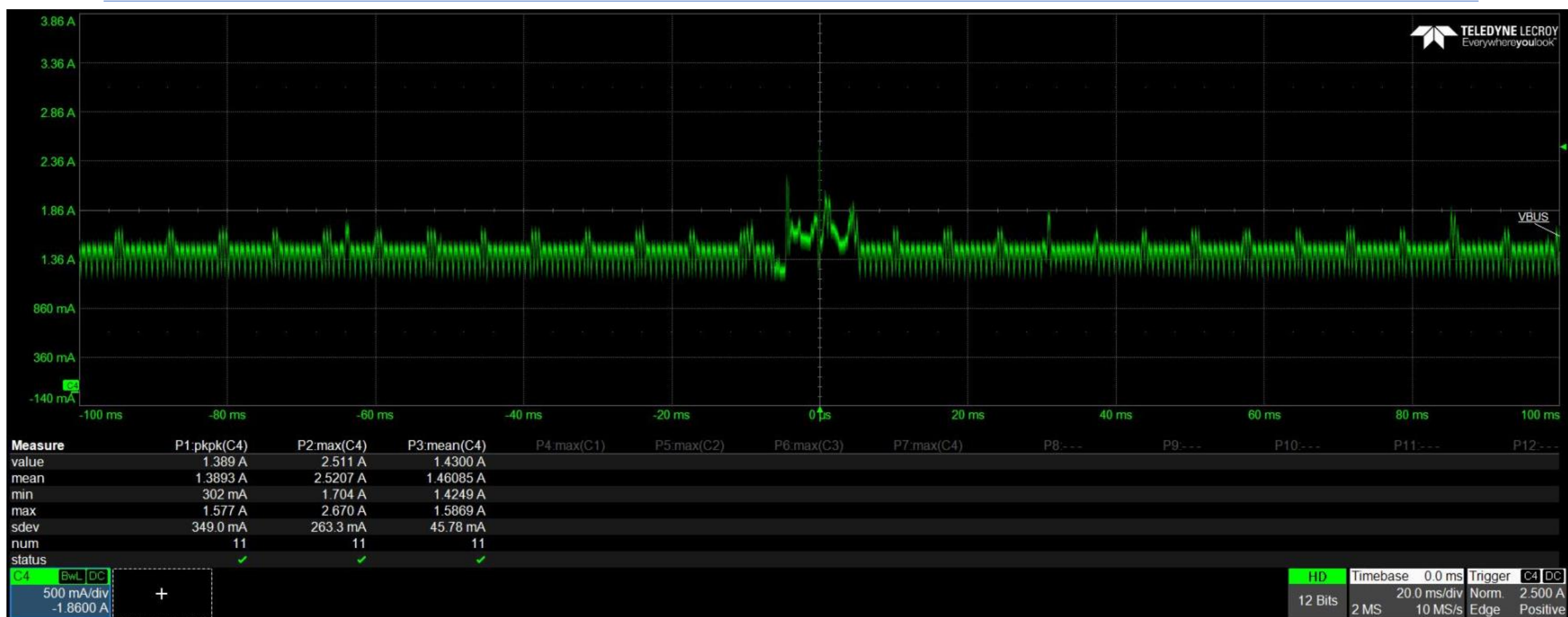
Norm. 2.500 A

Edge Positive

Added POSCAP 450UF/6.3V (3528/B2)20% TCJB157M006R0070 AVX ,
total capacity = 30.1uF ->**480.1uF** on the +3.3V power for M.2 SSD.
The Max peak current measured on VBUS (5V) : **2.663A** (SSD is 2TB)



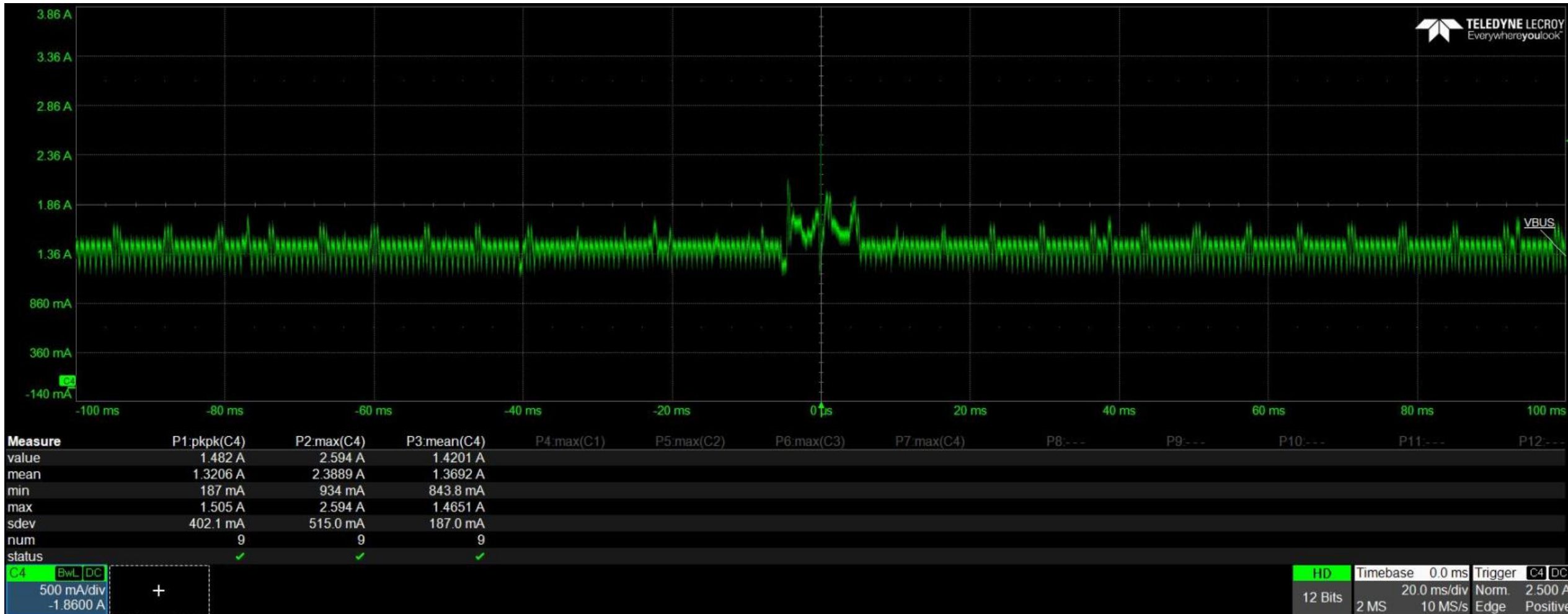
Improved USB4/TBT3 VBUS Current by Adding capacitance to **630.1uF** on +3.3V for M.2



Added POSCAP 600UF/6.3V (3528/B2)20% TCJB157M006R0070 AVX ,
total capacity = 30.1uF -> **630.1uF** on the +3.3V power for M.2 SSD.
The Max peak current measured on VBUS (5V) : **2.670A** (SSD is 2TB)



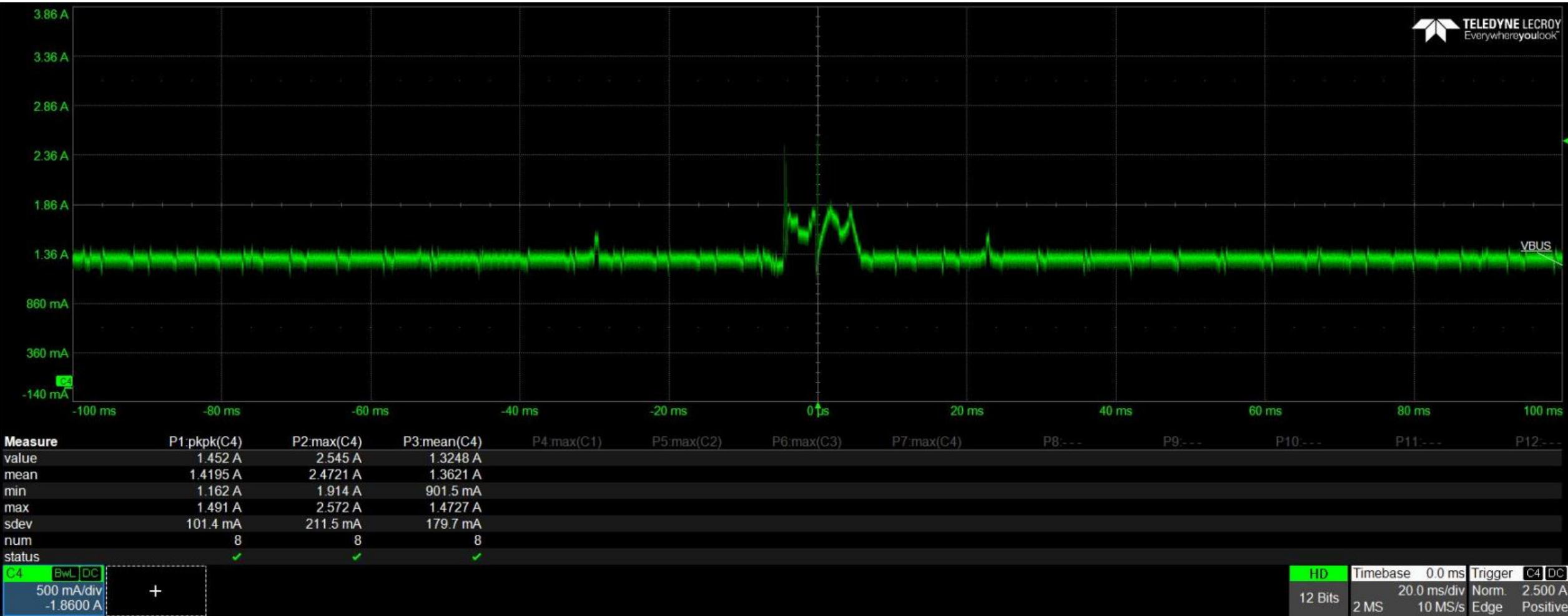
Improved USB4/TBT3 VBUS Current by Adding capacitance to **780.1uF** on +3.3V for M.2



Added POSCAP 750UF/6.3V (3528/B2)20% TCJB157M006R0070 AVX ,
total capacity = 30.1uF -> **780.1uF** on the +3.3V power for M.2 SSD.
The Max peak current measured on VBUS (5V) : **2.594A** (SSD is 2TB)



Improved USB4/TBT3 VBUS Current by Adding capacitance to **930.1uF** on +3.3V for M.2



Added POSCAP 880UF/6.3V (3528/B2)20% TCJB157M006R0070 AVX on location ,
total capacity = 30.1uF -> **930.1uF** on the +3.3V power for M.2 SSD.

The Max peak current measured on VBUS (5V) : **2.572A** (SSD is 2TB)



Summary of Improved USB4/TBT3 VBUS Current (SSD is Samsung 990PRO 2TB)

VBUS: 5V	Vbus peak current (A)
30uF	3.364
180uF	3.040
330uF	2.791
480uF	2.663
630uF	2.670
780uF	2.594
930uF	2.572

VBUS: 4V	Vbus peak current (A)
30uF	SSD lost
180uF	3.099
330uF	2.917
480uF	3.059
630uF	2.927
780uF	2.771
930uF	2.988



Summary of Improved USB4/TBT3 VBUS Current (SSD is ADATA Legend 920 4TB)

VBUS: 5V	Vbus peak current (A)
30uF	3.871
180uF	3.582
330uF	3.505
480uF	3.352
630uF	3.349
780uF	3.279
930uF	3.306



Summary of Improved USB4/TBT3 VBUS Current (SSD is Seagate FireCuda 530 4TB)

VBUS: 5V	Vbus peak current (A)
30uF	SSD Lost
180uF	SSD Lost
330uF	3.92
480uF	3.738
630uF	3.638
780uF	3.694
930uF	3.681



Thank you!